IT-SIMPLERA INSTITUTE

Network Administration Internship Program

WEEK 01 ASSIGNMENT REPORT

Networking Fundamentals & Network Simulation Tools

Issigned By	IT-SIMPLERA INSTITUTE
Assigned To	Mashal Hijab
Week	01
Submission Deadline	3 July 2026
Tools Used	Cisco Packet Tracer, GNS3, PuTTY

1. Introduction

This report documents the completion of the Week 1 internship task under the Network Administration Internship Program at IT-Simplera Institute. The objective of this task was to build a foundational understanding of computer networking concepts and to gain hands-on experience with two industry-standard network simulation platforms: Cisco Packet Tracer and GNS3. As part of the task, a simple Local Area Network (LAN) topology was designed and implemented in both tools, static IP addressing was configured on all end devices, and end-to-end connectivity was verified using the ping utility.

2. Networking Concepts Learned

The following core networking concepts were studied and applied during this task:

- Network Devices: A switch operates at Layer 2 and forwards frames based on learned MAC addresses, while a router operates at Layer 3 and forwards packets between different networks based on IP addresses.
- IP Addressing: Every device on a network requires a unique IP address for identification and communication. This task used private Class C addressing (192.168.1.0/24) with a subnet mask of 255.255.255.0.
- LAN (Local Area Network): A network confined to a small geographic area, such as an office or lab, where devices are directly connected through a switch.
- MAC vs IP Address: The MAC address is a fixed hardware identifier assigned to a network interface, whereas the IP address is a logical address that can be configured or reassigned.
- ICMP and the Ping Utility: Ping uses the Internet Control Message Protocol (ICMP) to send echo requests and measure round-trip time, making it a primary tool for verifying connectivity between two hosts.
- Static IP Configuration: Assigning IP addresses manually to each device rather than relying on an automatic (DHCP) service, which is useful in small, controlled lab environments.

3. Cisco Packet Tracer Interface Overview

Cisco Packet Tracer provides a graphical simulation environment for designing and testing network topologies. The main interface consists of a device selection panel on the bottom-left (grouped into routers, switches, end devices, and connections), a workspace/canvas in the center for placing and wiring devices, and a Physical/Logical toggle to switch between logical topology view and physical rack view. Devices are placed by dragging them onto the canvas, connected using the appropriate cable type, and configured through their Config or Desktop tabs. A Realtime/Simulation mode toggle in the bottom-right allows packets to be observed as they travel across the network.

4. GNS3 Interface Overview

GNS3 is a network emulation platform commonly used for more advanced, closer-to-real-world simulations. Its interface includes a device panel on the left (routers, switches, end devices/VPCS, and security devices), a central workspace for building the topology, a Topology Summary panel on the right listing all nodes and their console access details, and a Servers Summary panel showing the status of the local or virtual machine server running the simulation. Devices are started using the Start (play) control, and each device's console can be accessed directly for command-line configuration, similar to real network hardware.

5. Topology Design

An identical LAN topology was built in both platforms: one switch connected to three end devices (PCs), forming a simple star topology. Each PC was assigned a static IP address within the 192.168.1.0/24 subnet.

Device	IP Address	Subnet Mask
PC1 / PC0	192.168.1.10	255.255.255.0
PC2 / PC1	192.168.1.11	255.255.255.0
PC3 / PC2	192.168.1.12	255.255.255.0

5.1 Cisco Packet Tracer Topology

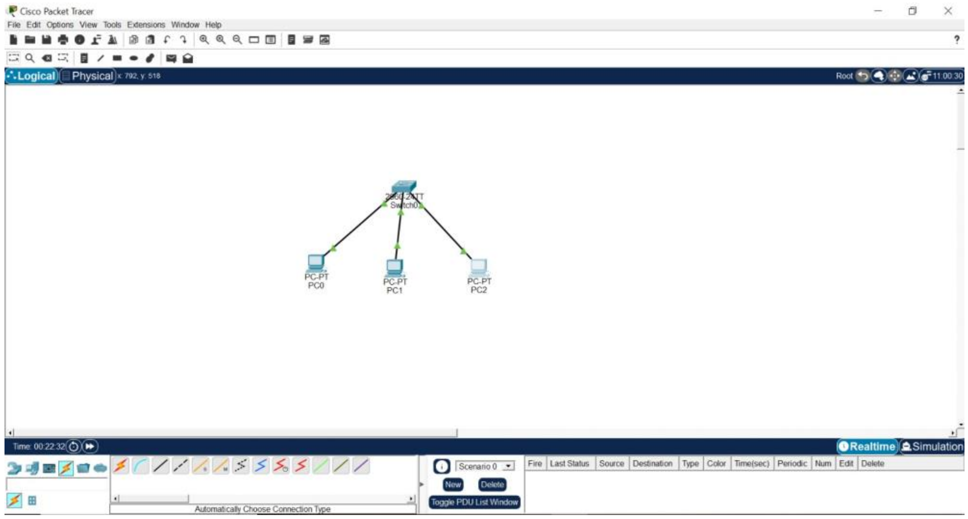

Figure 1: LAN topology in Cisco Packet Tracer — one switch connected to PC0, PC1, and PC2.

5.2 GNS3 Topology

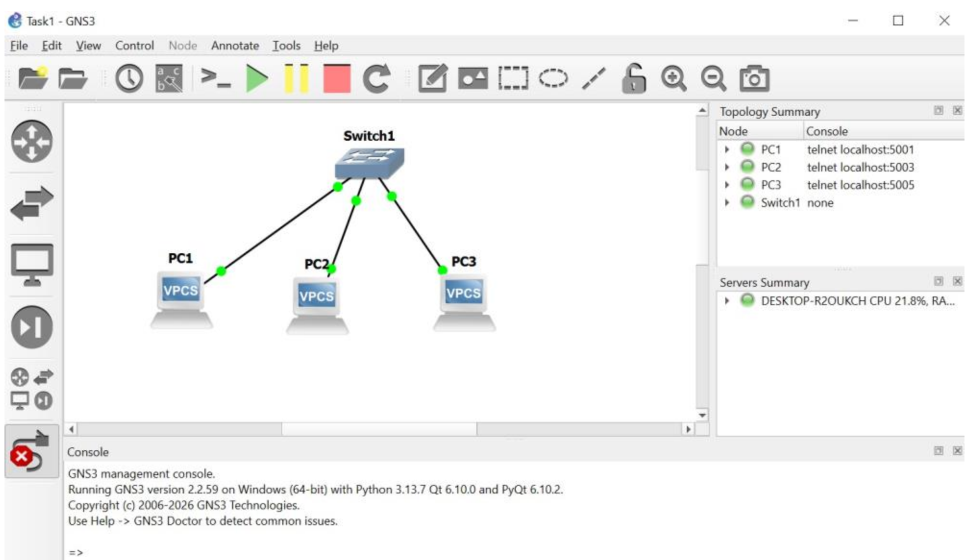

Figure 2: LAN topology in GNS3 — Switch1 connected to PC1, PC2, and PC3 (VPCS nodes).

6. Ping Test Results

Connectivity between all devices was verified using the ping command in both platforms. All tests returned successful replies with 0% packet loss, confirming that the LAN topology, IP addressing, and switch forwarding were configured correctly.

6.1 Packet Tracer Ping Test

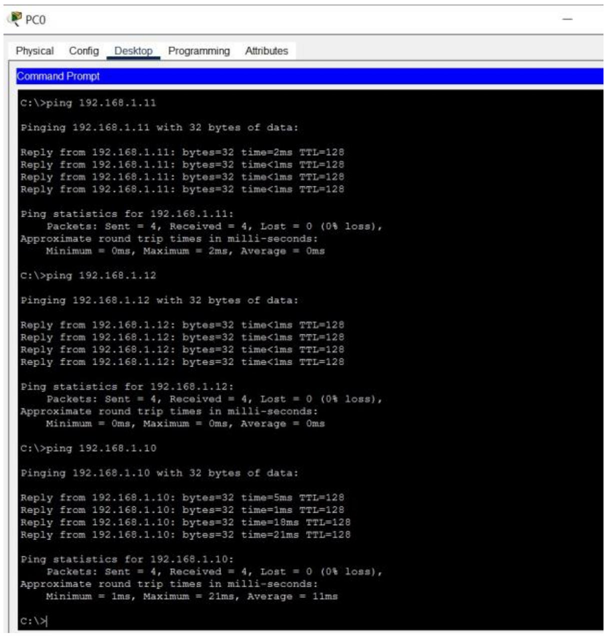

Figure 3: Successful ping results from PC0 to 192.168.1.11, 192.168.1.12, and 192.168.1.10 (0% loss).

6.2 GNS3 Ping Test

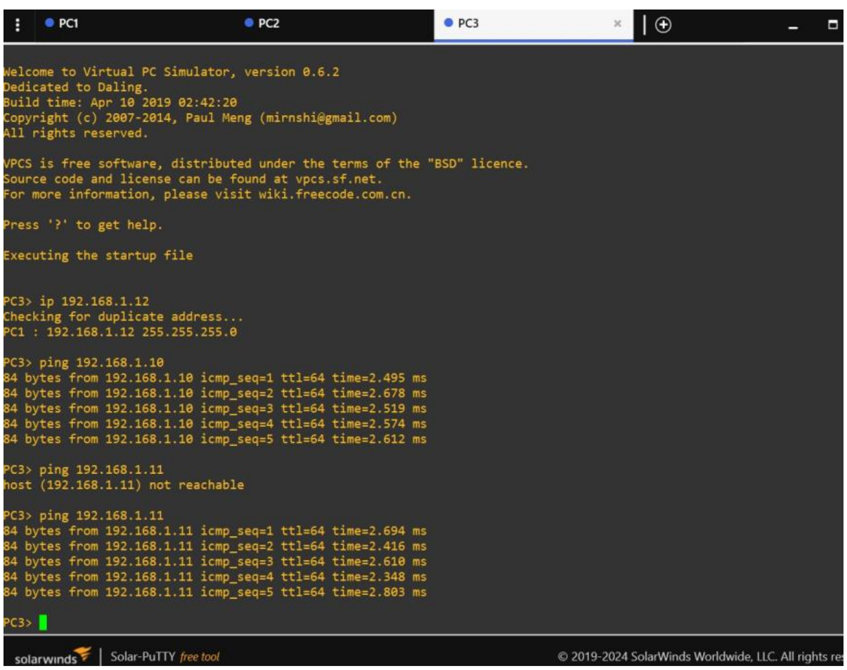

Figure 4: Successful ping results from PC3 to PC1 (192.168.1.10) and PC2 (192.168.1.11) via the VPCS console.

7. Challenges Faced

- Initial connectivity issue in GNS3 where a ping attempt to 192.168.1.11 returned "host not reachable" before the destination VPC's IP configuration had been fully applied; the issue resolved after re-attempting the ping once the device was correctly configured.
- Understanding the difference between GNS3's VPCS console (command-line based, telnet access) and Packet Tracer's Desktop-based Command Prompt required some adjustment, as the two tools use different interfaces for the same ping/IP tasks.
- Setting up the GNS3 environment (VM/local server integration) required more initial configuration compared to the more self-contained Packet Tracer application.

8. Conclusion

This task provided practical, hands-on exposure to core networking fundamentals and to two widely used network simulation tools. By building the same LAN topology in both Cisco Packet Tracer and GNS3, configuring static IP addressing, and verifying connectivity through successful ping tests, a solid foundation was established for more advanced networking tasks in the coming weeks, including router configuration, VLANs, and dynamic routing protocols.